Probability :

- An experiment → it is the process by which an observation (or measurement) is obtained
- trial → doing experiment for one time.
- Outcomes of a trial are determined to be equally likely.
- The collection of all possible outcomes of a random experiment → sample space (Ω), (S)

Let's say we have to build a password with 2 letters and 1 number. The possible ways are —

→ $26 \cdot 26 \cdot 10 = 6760$ ways of combinations.

If a dice and a coin is thrown at the same time ⇒

S or Ω = {1H 1T 2H 2T 3H 3T 4H 4T 5H 5T 6H 6T}

Permutation :

Order is important, no repetition

$$nP_r = \frac{n!}{(n-r)!}$$

Combination :

Order is not important repetition is allowed

$$nC_r = \frac{n!}{(n-r)!\,r!}$$

- Also permutation with ~~reputation~~ repitition formula :

$$\boxed{\frac{nP_n}{P!\,q!}}$$

- Combination with repititon formula :

$$C(n,r) = \frac{(n+r-1)!}{r!\,(n-1)!}$$

ex:

A race consists of 10 runners, one can receive gold, silver, broze medal in possible ways →

⇒ here $P_{(10,3)} = \frac{10P_3}{(10-3)!} = \frac{10!}{(10-3)!}$

$= 10 \times 9 \times 8 = 720$ ways.

A club has 10 members. A 3 person committee can be formed in possible ways of —

$^{10}C_3 = \frac{10!}{(10-3)!\,3!} = 10 \times 3 \times 4 = 120$

For making a cake 3 ingredients, 3 icings, 15 topping the possible ways —

$3 \times 3 \times 15$ ways.

5 books can be arranged in a shelf —

$^5P_5 = 5! = 120$ ways

A 4 digit pin can be created —

with repitition $= 10 \times 10 \times 10 \times 10 = 10000$ ways

without " $= 10 \times 9 \times 8 \times 7 = 5040$ "

There is 5 consonants and 3 vowels how many ~~ma~~ word of 3 consonants and 2 vowels can be ~~perfo~~ formed ?

$\rightarrow$ $5P_3 \times 3P_2 = 360$

~~Phone~~ PHONE $\rightarrow$ Organize the words where vowels can be together.

$\rightarrow$ $4! \times 2! = 48$ ways.